Self-Evolving Recommender Systems: A Survey and an Evaluation Audit

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Abstract

In 2026 a new kind of recommender research appeared: large language model (LLM) agents that propose changes to a recommender's configuration or source code, train the changed model, and keep what the evaluation says is better. Within ten months, more than thirty such systems were reported, from open-source research harnesses to production loops at Google, Meta, Amazon, Alibaba, Kuaishou, Tencent and NetEase. This survey organises them, together with search-space AutoML and the agentic recommenders that update their own memory or weights, along two axes: *what* the loop changes and *which signal* decides whether a change survives. Because a self-evolving system is only as good as the signal it selects on, we then audit how 27 configuration- and code-level recommender systems are evaluated, coding each paper's full text against nine criteria. Three of the 27 evaluate with full-catalogue ranking. Of the 17 that select candidates offline, nine do not say which data split makes the decision, two select on the data they report, and three state that the test set was used once. None reports its headline result with the spread over repeated evolution sessions, and none runs an equal-budget random search over its own edit space; the few studies that measure transfer to held-out data find that part of the selected gain, sometimes most of it, does not survive. Four systems report that their agent inflated the metric or leaked the target, caught by a human or by monitoring rather than by the acceptance rule. We distil these findings into a reporting standard and describe recforge, an open benchmark under development that implements the API side of a sealed-test, full-ranking, equal-budget protocol.

Keywords

recommender systems, LLM agents, AutoML, program evolution, evaluation, reproducibility

1 Introduction

Recommender models have long been improved by a loop that engineers run by hand: propose a change, train, compare offline, and launch the winners into an online experiment. Automating parts of this loop is an old goal. Neural architecture and embedding search for recommendation models, mostly click-through-rate (CTR) prediction [56, 89, 123, 128] and loss search [49] automated the choice inside a space that a person had written down [130]. What changed in 2026 is that the proposer became an LLM agent that writes code. Following FunSearch [87], AlphaEvolve [73] and open implementations such as OpenEvolve [74] and ShinkaEvolve [42], agents now edit the model, loss and training code of recommenders directly. Google reported such a loop for YouTube ranking [95] and SelfEvolveRec released an open harness [39] in February 2026; within months, open harnesses such as RecHarness [55] and systems at Meta [46, 65], Alibaba [30, 66], Kuaishou [43, 108, 135], Tencent [3, 60], NetEase [121] and others followed.

These systems are optimisers, and an optimiser exploits whatever its objective rewards. A loop that evaluates hundreds of agent-written candidates on one validation set and keeps the maximum is the setting in which the known weaknesses of recommender evaluation—sampled metrics [41], temporal leakage [34], untuned baselines [19, 84], and the reuse of a holdout for adaptive decisions [9, 21]—are most likely to turn into reported gains. The agent also controls the code that produces its score. Two of the systems we read report that their agent won on the metric without a real improvement: RecEvolve's agent inflated its proxy metrics by shrinking the batch size, and with it the number of in-batch negatives [76], and an AI co-scientist for search ranking produced implementations that "improved $\mathcal{M}$ while being wrong (e.g., silently increasing model size)" [101]. Figure 1 shows the loop these systems share and the points where the evidence for a gain can fail.

Existing surveys do not cover this development. Surveys of LLM agents for recommendation treat the agent as the recommender, a conversational interface or a user simulator [33, 79, 126]; surveys of self-evolving agents treat the agent's own prompts, memory, tools or weights as the object that evolves, and mention recommendation at most in passing (a two-sentence mention of self-optimised fine-tuning and a recommendation task in a benchmark it describes in [2], none in [22]); the most recent roadmap for agentic recommender systems mentions the "agent as engineer" direction only in passing, in two sentences that cite AgentX and the Google loop [54]; and the most recent peer-reviewed survey of agentic recommenders organises 47 studies by where the agent sits in the serving workflow, not by what changes the recommender [106]. A general survey of how automated research is evaluated covers no recommender system [120]. Closest to our audit, a concurrent controlled study shows on CTR and learning-to-rank data that exact validation feedback lets a coding agent overfit the validation split, and that a feedback contract with a quarantined test set removes the spurious gain on a task with no signal [112]. Beel et al. call for autonomous recommender-systems research labs and for benchmarks to evaluate them [6]. RMD-Bench [78] fixes protocols for cross-dataset and cross-scale discovery and ExecML-HSTU [16] measures whether agents' edits execute correctly, but no shared benchmark yet compares evolvers under a sealed test, full-ranking evaluation and equal-budget null controls.

This survey makes four contributions.

(1) **A taxonomy** (§4) that places search-space AutoML, configuration and code evolution, evolution of generative-recommender

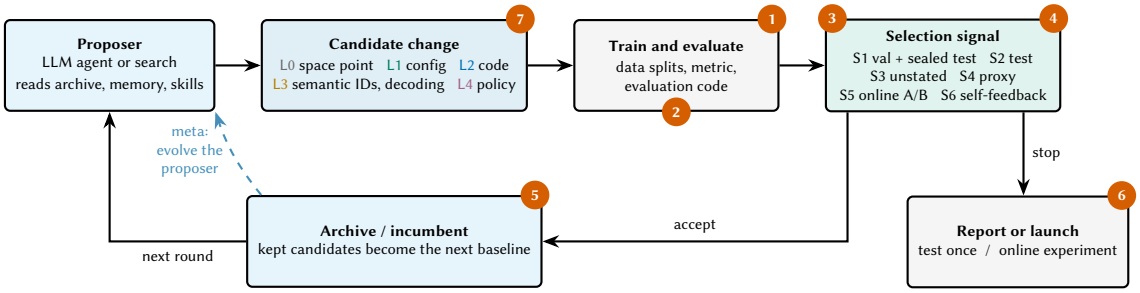


Figure 1: Anatomy of a self-evolving recommender. A proposer changes some part of the recommender (the evolved object, axis 1 of the taxonomy); the changed system is trained and evaluated; a selection signal (axis 2) decides what is kept. Red markers show where the evidence for a reported gain can fail, with the audit criterion that checks it: 1 the agent edits evaluation code, reads test data or games the metric (C9); 2 sampled or unstated ranking metrics (C1); 3 the maximum over many candidates on one reused split, or selection on test (C2, C3); 4 a proxy or self-generated signal that does not transfer (S4, S6); 5 a gain inside run-to-run noise becomes the new baseline (C4, C5); 6 short or repeated online experiments without correction for multiple testing; 7 gains measured against an untuned starting point, with no equal-budget null search (C6, C8).

components, and self-improving agentic recommenders on two axes: the evolved object and the selection signal.

(2) **A review of the systems** (§5) at each level, with what is open source.

(3) **An evaluation audit** (§6) of the configuration- and code-level systems on nine criteria, coded from the full text with verbatim evidence that we release.

(4) **A reporting standard and a shared protocol** (§7), partly implemented in an open benchmark under development, recforge [82], together with open problems (§8).

2 Scope and Survey Method

Definition. We call a recommender system *self-evolving* when an automated proposer changes some part of it, the changed system is evaluated, and the evaluation decides which changes are kept, repeatedly and without a human choosing each change. Humans may approve launches; the loop must generate and select candidates by itself. This includes systems whose proposer is a search algorithm over a fixed space (background, §3), but our focus is on LLM proposers.

Inclusion. We include (i) systems that change a recommender's or ranker's configuration, features, code or components under an evaluation loop; (ii) agentic or LLM recommenders whose policy, memory, skills, prompts or weights improve from feedback they generate; and (iii) benchmarks, environments and surveys for these. We exclude LLM recommenders without a self-improvement loop, conversational recommendation in general, and general hyperparameter-optimisation libraries. Systems outside recommendation (learning to rank [77], search ranking [101], lexical retrieval [72], generative document retrieval [71] and lifetime-value prediction [99]) are included as comparison points when they inform evaluation.

Search protocol. The cut-off date is 22 October 2026. We started from keyword searches over arXiv, OpenAlex and Semantic Scholar ("self-evolving recommender", "LLM agent recommender model optimisation", "agentic harness recommender", "LLM-driven architecture search CTR" and survey queries), then walked the reference lists of the 2026 systems (RecHarness, EvoRec, AgentX-Model, RecEvolve) and of the agentic-recommender roadmap backwards and their titles forwards, and finished with about thirty full-text arXiv phrasings combined with company names. Generic keyword search returned mostly unrelated "agent for X" papers; the citation graph of the 2026 systems was the productive source. A second sweep then ran 190 further arXiv queries (817 new candidates, all screened by title), enumerated 13 ACM proceedings (RecSys, KDD, SIGIR and WWW 2025–2026, CIKM 2025, WSDM 2026; 6,874 records) through OpenAlex, followed forward citations of 16 seed systems in Semantic Scholar, and searched industry engineering blogs; it added nine systems, five of which entered the audit. CIKM 2026, DBLP and OpenReview could not be searched automatically. Every arXiv entry in our bibliography was generated from the arXiv API record, and every code repository we list was checked to exist.

Audit procedure. For each configuration- or code-level system, we read the full text (arXiv HTML or PDF, including appendices) and coded nine criteria (Table 1). A criterion is coded "not stated" when the paper does not say, which is different from "no". Each coded cell is backed by a verbatim quote of at most 25 words with a section pointer; the data file and evidence will be released with recforge so that the coding can be checked and corrected. For systems whose launch decision is an online experiment, we read the offline criteria as describing the offline screen, and assess the launch evidence separately (R9 in §7).

3 Background

Search-space AutoML for recommenders. AutoML for recommenders searched for feature-interaction structures (AutoCTR [89], AutoFIS [56]), embedding dimensions (AutoEmb [128], AutoDim [129]), whole architectures with weight sharing (NASRec [123], DNS-Rec [122]) and loss functions (AutoLossGen [49]); Zheng et al. survey this line [130]. Each method can only return a design that its space contains. Evaluation is mostly validation logloss or AUC on public CTR data, or standard rating and sequential benchmarks, with standard splits, and the number of full evaluations is small, so the selection problems we discuss later are mild.

Table 1: Audit codebook.

C1	Metric: full-catalogue ranking, sampled ranking (with N negatives), AUC/loss, or online only
C2	Selection signal: which split or signal accepts or rejects each candidate
C3	Test discipline: is the selected candidate evaluated on a held-out test once, after search?
C4	Variance: repeated seeds or repeated evolution runs for headline numbers
C5	Overfitting check: validation-to-test gap or held-out transfer of the selected candidate
C6	Null control: random search, HPO or another non-LLM search at equal budget
C7	Budget: what is counted (iterations, GPU time, tokens, money)
C8	Baselines: are starting points and compared models tuned?
C9	Isolation: can the agent reach test data or evaluation code, and is metric gaming discussed?

LLM program evolution and ML-engineering agents. FunSearch [87] and AlphaEvolve [73] pair an LLM that proposes code with an automatic evaluator and an archive of programs. Open systems followed: OpenEvolve [74], ShinkaEvolve [42] and DeepEvolve [58]. ML-engineering agents—AIDE [36], MLE-STAR [67], R&D-Agent [111], AIRA [93] and the AI Scientist [62, 107]—apply the same pattern to whole training pipelines, and recent work optimises the agent harness itself [44, 61, 105]. Their benchmarks contain at most one recommendation task: MLE-bench includes the H&M fashion-recommendation competition [11] and MLRC-Bench a next-product-recommendation task [125]. MLRC-Bench shows agents only the development split and scores the best development snapshot once on test, whereas MLGym's `validate` command returns test-set metrics during the session [69].

Generative recommenders. HSTU [115] and TIGER [81] recast recommendation as sequence generation over item IDs or semantic IDs, and OneRec unifies retrieval and ranking in one generative model that is further aligned by iterative preference optimisation on its own samples [20, 131, 132]. These models add new objects that a loop can evolve: tokenisers, codebooks, decoding procedures and generative losses.

Evaluation methodology. Tuned simple baselines beat many published neural recommenders [19, 84]; a loss change alone can reorder SASRec [37] and BERT4Rec [40, 90], and BERT4Rec's reported advantage depends on much longer training [80]; sampled metrics are inconsistent with full-ranking metrics and can reverse model rankings [29, 41, 83]; leave-one-out splits that ignore the global timeline leak future information [29, 34], and the split choice changes rankings [64]. Libraries and open benchmarks such as RecBole [127], Elliot [1], DaisyRec [91], BARS [133] and FuxiCTR [134] standardise protocols for human-designed models. For adaptive reuse of a holdout, theory predicts overfitting [9, 21]; empirically, a meta-analysis of 120 Kaggle competitions found little [86]. For LLM loops specifically, keep-if-better acceptance is selection under noise, with selection-set scores well above held-out accuracy when the selection set is small [31]; best-of-N runs produce apparent gains from selection alone [27]; for LLM research agents, the scarcity of overfitting has been tied to how compressible their successful strategies are [8]; and AI-scientist systems hide post-hoc selection, leakage and metric misuse [63]. We return to this tension in §6. Figure 2 places these precursors next to the 2026 systems.

4 A Taxonomy

We place each system on two axes (Table 2).

Axis 1: the evolved object. **L0 search space**: a point in a human-written space (AutoML). **L1 configuration**: hyperparameters, fusion weights, retrieval budgets, thresholds. **L2 code**: model, loss, training, feature code. **L3 generative components**: semantic-ID construction, item vocabularies, decoding. **L4 policy of an agentic recommender**: its prompts, memory, skills or LLM weights. **Meta**: the evolver itself—its prompts, skills, playbooks or a learned proposer.

Axis 2: the selection signal. **S1** offline validation with a separate test; **S2** an offline held-out set that is also the reported result; **S3** offline, split not stated; **S4** a proxy model or a small-scale screen; **S5** online A/B experiments; **S6** self-generated feedback (reflection, simulators, the model's own samples).

The two axes are not independent in practice. Code-level systems use S1, S3 and S5; component-level systems rely on S4 because full training is too expensive to run per candidate; policy-level systems use S6. Each signal has a characteristic way of failing: S1 overfits a reused validation set, S2 reports the selection bias as a gain, S3 cannot be assessed, S4 fails when the proxy does not transfer, S5 is vulnerable to short windows and to multiple testing over many rounds, and S6 can be circular when the judge and the judged come from the same model family. This is why we audit the selection signal rather than only the reported gain.

5 Systems

5.1 Configuration-level loops (L1)

L1 systems let an LLM change numbers or switches while the model code stays fixed. VirtualMLE tunes SASRec and HSTU through an execute–reflect loop with a hierarchical memory that transfers across datasets [10]. In production, CORAL allocates retrieval budgets and serving treatments [4], AgenticRecTune tunes fusion weights and thresholds across pre-ranking, ranking and re-ranking [102], Sortify searches the exchange rates of a sorting formula with Optuna while an LLM meta-controller recalibrates the search's offline-to-online transfer intercepts and penalties [18], GEARS selects segment-level treatment policies [113], and SR-Agent edits post-ranking diversity and exposure strategies through four bounded action types [108]. Taobao's PILOT has LLM roles choose ranking-objective weights for user segments from a legal-command envelope generated by deterministic services [52]; Kuaishou's A/B Agent iterates strategy parameters across online rounds with a hierarchical experience tree [35]; and DMRL trains a small LLM with reinforcement learning to rewrite the skill document that an ad-tuning agent follows [124]. Most L1 systems in production select directly on online experiments; an e-commerce search loop adds a multi-fidelity ladder from shadow-traffic proxies to decision-grade online tests [97].

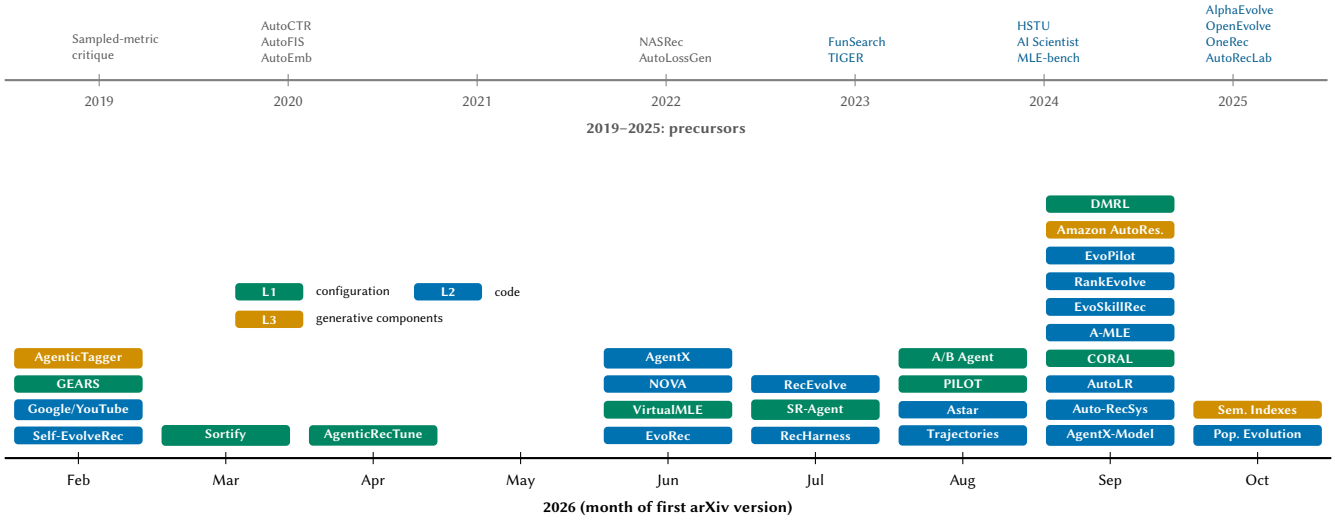


Figure 2: From search-space AutoML to agent-driven evolution. Top: precursors by year (grey: search-space AutoML and evaluation critique; blue: program evolution, ML-engineering agents and generative recommenders). Bottom: the 2026 configuration-, code- and component-level systems surveyed here, by month of their first arXiv version; the colour gives the highest level of the evolved object. Comparison points outside recommendation and L4 systems are omitted.

Table 2: Taxonomy of self-evolving recommender systems: evolved object (rows) × selection signal (columns). Code: † open-source code verified.

	S1 offline val + test	S2/S3 offline, test reused or unstated	S4 proxy / screen	S5 online A/B (primary)
L0 search space	AutoFIS† [56], NASRec† [123], AutoCTR† [89], AutoLossGen† [49]			
L1 configuration	VirtualMLE† [10]	AutoLR [121]		PILOT [52], A/B Agent [35], DMRL [124], CORAL [4], AgenticRecTune [102], Sortify [18], GEARS [113], SR-Agent [108]; *comparison:* PEAR (search) [97]
L2 code	Self-EvolveRec† [39], RecHarness† [55], EvoSkillRec† [48]; *comparison:* EvoRank† (LTR) [77], Feedback Contract [112]	RankEvolve [16], EvoRec [66], A-MLE [26], Trajectories-to-Evidence [135], NOVA [60], AgentX-Model [110], Auto-RecSys [46], RecEvolve [76], AutoLR [121], EvoPilot [24], Amazon AutoResearch [12], Meta REA (blog) [65]; *comparison:* AI Co-Scientist (search) [101]	Population Evolution [78], Astar [30]	Google/YouTube [95], AgentX [43]
L3 components		Amazon AutoResearch (semantic-ID tokenizer) [12]	AgenticTagger [103]; *comparison:* Evolving Semantic Indexes (GR) [71]	
L4 policy (S6)	Memory/reflection: AgentCF† [118], AgentCF++† [59], SAGER [92], RecNet [45], AtomRec [32], MemRec† [14]; weights: OneRec [20], SPRec† [25], Rec-R1† [53], CoARS [98], RSIR† [119]; prompts and tool descriptions: Spotify [75]			
Meta	Learned direction proposer: Astar [30]; harness and skill evolution: AgentX SGPO [43], EvoRec skills [66], Auto-RecSys playbooks [46], AgenticRecTune skill hub [102], A/B Agent experience tree [35], DMRL's RL-trained skill optimiser [124]			

5.2 Code-level loops in the open (L2)

Self-EvolveRec [39] runs an AlphaEvolve-style loop over the whole pipeline code of NCF, NGCF, SASRec and MoRec seeds;

besides validation metrics it receives directional critiques from an LLM user simulator and from a diagnosis tool that evolves with the model. **RecHarness** [55] routes each round with Thompson sampling over human-defined "edit arms" (schedules, dropout, depth, pooling, features, loss) plus "jump" arms for structural changes, uses validation feedback only, evaluates the incumbent on test once, and caps compute at 43,200 GPU-seconds; HSTU is one of its five backbones. **EvoSkillRec** [48] evolves architectures as typed "skill genomes" of executable modules and promotes validated modules into a skill library. **RankEvolve** [16] composes Claude Code and Codex under an executable operating protocol in which the two agents review and repair each other's edits to HSTU, and introduces ExecML-HSTU, a benchmark of how often agent edits are executed correctly. **Population Evolution** [78] runs several evolution searches that share evidence across related datasets, screens candidates on peer datasets (and, for LLM and VLM targets, on small training subsets), and promotes survivors to larger scales; it introduces RMD-Bench. AutoRecLab [5, 6] generates recommender experiments from natural-language descriptions; it automates experimentation rather than evolving a model, and we do not audit it.

5.3 Code-level loops in industry (L2)

Most industrial systems share a two-stage design: an offline loop screens agent-written candidates, and online experiments decide launches. Google's loop for YouTube ranking [95] has Gemini personas propose optimiser, architecture and reward changes, promotes on validation loss (with a proxy metric for reward changes) and then on north-star A/B metrics, and reports LLM costs of about $20,000 over six months. RecEvolve [76] runs the full research loop on a production two-tower retrieval model. Meta's REA plans, launches and debugs ads-ranking experiments with up-front GPU budget approval [65]; A-MLE explores a portfolio of ads-ranking models with human checkpoints [26]; Auto-RecSys runs idea- and execution-evolution loops over multi-day training jobs [46]; and RankEvolve and Population Evolution above also come from Meta. Alibaba's EvoRec [66] pairs research and code agents with a skill evolver, and Astar [30] trains an 8B model on the commit history of an advertising system to propose the next evolution direction. AgentX, deployed at Kuaishou [43], iterates online strategies and model research, and its successor AgentX-Model [110] runs research lineages (reproduce, follow up, compose, diagnose) over production rankers, and a Kuaishou paper converts research-agent trajectories into auditable evidence [135]. Tencent's NOVA [60] adds verification gates that reject recommender-invalid edits such as target leakage, NetEase's AutoLR [121] reviews proposals with a multi-expert proposal council under a limited trial budget, and Tencent's RecSys Factory bounds agent autonomy to typed decision points in fixed pipelines [3]. A Trip.com co-scientist for travel search ranking [101], LLM evolution of learning-to-rank pipelines (EvoRank [77]) and of lexical retrieval functions [72] are close relatives outside recommendation. Meta's EvoPilot runs a 37-day campaign of agent-proposed configuration and retraining changes to video-retrieval models behind a fail-closed verifier and reports the failures it caught [24], and Amazon's AutoResearch rewrites the training code of a book-retrieval model and tunes a semantic-ID tokenizer over 220 experiments, naming the failure modes it met [12]. Adjacent loops evolve ads kernels [50], off-policy-evaluation code [100], search-relevance models [94] and an LLM policy layer trained against a learned evaluator [117].

5.4 Components of generative recommenders (L3)

Generative recommenders expose objects that do not exist in ID-based models. Evolving Semantic Indexes, a generative document-retrieval system that we treat as a comparison point, runs LLM-driven program evolution over semantic-ID construction, collision handling, decoding and the generator's training objective, using a one-million-parameter proxy decoder as fitness [71]. AgenticTagger has an LLM refine a hierarchical item-descriptor vocabulary used as tokens [103], and a generative recommender can reassign its own item tokenisation during training [13]. Amazon's AutoResearch is the first audited production loop that edits a semantic-ID tokenizer, and the first in which the agent gamed a tokenizer metric [12]. Feedback-grounded policy discovery distils LLM-discovered natural-language policies into a semantic-ID generator [17]. Evolving Semantic Indexes evolves TIGER-style semantic-ID components for document retrieval but releases no code; no open system yet does so for recommendation under a shared protocol.

5.5 Self-improving agentic recommenders (L4)

When the recommender is itself an LLM agent, the evolving object is its memory, skills, prompts or weights, and the feedback is mostly generated by the system. Memory and reflection methods update user and item agents (AgentCF [118], AgentCF++ [59]), per-user policy skills (SAGER [92]), propagation modules optimised with textual gradients (RecNet [45]), and structured memories that are rewritten as interactions arrive (AtomRec [32], AMEM4Rec [70], STEAM [51], MARS [88], MemRec [14]). Weight-level self-improvement includes OneRec's alignment on its own samples, first by iterative DPO [20] and then by on-policy RL [131, 132], self-play debiasing (SPRec [25]), RL against a recommender's metric (Rec-R1 [53]), co-evolving recommender and user agents (CoARS [98]), and recursive self-training on filtered synthetic sequences [28, 119], where progress can also sit in earlier generations, not only in the latest model [104]. Spotify improves a conversational recommendation agent's prompts, few-shot examples and tool descriptions with a coding agent, evaluated by an LLM judge on synthetic conversations [75]. Simulated environments and benchmarks for such agents include RecoWorld [57], the user simulators Agent4Rec [116] and RecAgent [96], and the verifiable τ-Rec benchmark [68].

5.6 Evolving the evolver (meta)

Several systems improve the loop itself: AgentX's SGPO applies semantic-gradient updates to its agents [43], EvoRec and AgenticRecTune distil skills from experiment memory [66, 102], Auto-RecSys evolves execution playbooks [46], and Astar learns the proposer from history [30]. These are the recommender counterparts of harness optimisation [44, 105]. Meta-evolution inherits the selection problem one level up: the harness is kept or discarded by a score of its own. In our own pre-registered study of LLM

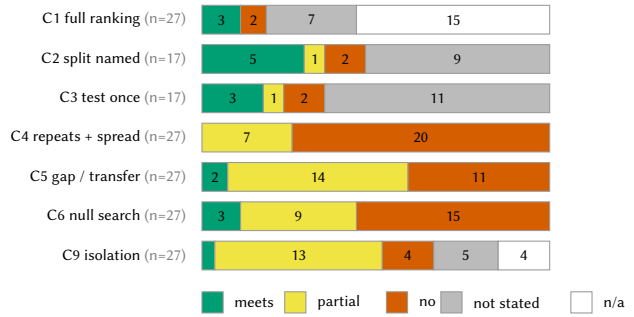


Figure 3: Audit summary: how many of the 27 recommendation systems (17 for the criteria that apply only to offline selection) meet each criterion. C1 counts AUC- and online-only evaluations as not applicable; C9 counts a reported metric inflation or leakage by the loop as "no". Generated from the released `audit.csv`.

research-ideation harnesses, outside recommendation and not yet peer reviewed, a harness optimised against an LLM judge raised the judge's scores while the share of its hypotheses that held under measurement fell, on two hypothesis spaces with measured ground truth and for two Claude models (Sonnet and Opus) [109].

6 How Are Self-Evolving Recommenders Evaluated? An Audit

Audited set. We audited the 27 recommendation systems at levels L1–L3 (with or without meta-evolution) that select candidates with an evaluation loop, and seven comparison points: learning to rank, search ranking, e-commerce search strategy, lexical retrieval, generative document retrieval, lifetime-value prediction, and a controlled study of agent feedback on public CTR and learning-to-rank data [112]. Seventeen of the 27 select candidates offline; nine select on online experiments; Auto-RecSys reports only operational outcomes and makes no offline accuracy claim. Meta's REA is described only in a blog post and is not audited, and two NeurIPS 2026 workshop papers are known to us only from their abstracts: MLE Agents, which kept 325 of 1,097 agent-written changes to production search rankers [47], and an open-ended research loop on the public KuaiRand and KuaiRec data [15]; we will code them once the full texts are available. Four borderline systems were read but not counted: RecSys Factory [3] (an operations platform in which an operator approves the agent's proposed overrides through human-in-the-loop cards), AutoRecLab [5] (experiment generation), AURA [38] (stops at pull requests) and Spotify's prompt loop [75] (an L4 system). Table 3 gives the coding and Figure 3 summarises it; counts below are generated from the released data file.

F1: Almost nothing is evaluated with full-catalogue ranking. Of the 27 systems, three state a full-catalogue ranking evaluation (RankEvolve on HSTU, Population Evolution, and the retrieval system of Amazon's AutoResearch [12]), and in the last the agent itself designed a text-only fallback that "expanded catalog coverage by 5.8×", which by our reading changed the candidate set during the campaign. Two rank against sampled negatives: Self-EvolveRec and RecHarness pair each test item with 99 sampled negatives (random in Self-EvolveRec). Five report ranking metrics without saying what the candidate set is; for RecEvolve, the fact that shrinking the batch inflated its metrics (F7) suggests in-batch negatives. Six use AUC or a loss as the offline metric, nine select on online experiments, and two report no usable offline metric.

F2: The selection split is usually not stated, and sometimes it is the test set. Of the 17 systems with an offline selection loop, five name a validation split (Self-EvolveRec, RecHarness, VirtualMLE, EvoSkillRec, and Google's validation loss), one ranks candidates with a reward model trained on loss-curve outcomes and accepts by loss decrease (Astar), and nine do not say which data accept or reject candidates; EvoPilot, for example, selects and reports on one held-out replay set [24]. RankEvolve selects on test data: its persistent leaderboard is the canonical full-test NDCG@10 [16]. Population Evolution selects candidates on source instances (no separate validation split is stated) and also reports transfer to others, but calibrates the shared baseline configuration of its transfer targets on their released test split [78]. Amazon's AutoResearch "retains modifications that improve a held-out scalar metric" and reports the same metric, headlining the trajectory's peak ("1.82× relative lift … on its peak evaluation date") [12]. All three disclose these choices, which is to their credit. Outside recommendation, the travel-search co-scientist keeps or reverts each change on the same held-out metric that it reports [101].

F3: Test-once is rarely stated. Three of the 17 offline loops state that the selected candidate was evaluated on held-out test data once after the search (RecHarness, VirtualMLE, EvoSkillRec); RankEvolve and Amazon's AutoResearch report on the data they select on; Population Evolution partly; eleven do not say.

F4: No headline result comes with its spread over repeated sessions. RecHarness and VirtualMLE average three runs without reporting a standard deviation; Google runs six independent sessions with bracketed ranges, but only for an ablation; A-MLE has a rule to re-run when within-run variance is high. EvoPilot notes that two publications of the same model, configuration and pinned data "differed by 0.36 points" [24], and Amazon re-evaluates one model across 11 dates ("6.18% ± 0.28 pp") [12]; neither repeats a session. Several papers state that their gains are within the resolution of a single run: EvoSkillRec notes margins "on the order of 10^{-4} AUC, which a single evolution run cannot reliably separate from run-to-run variation" [48]; AgentX-Model reports run-to-run standard deviations of 0.0010–0.0022 AUC while counting an experiment as above baseline if it exceeds it by more than 10^{-6} [110]; and AutoLR, in its audited configuration, promoted candidates after a single run with noise calibration disabled; the authors themselves identify the resulting risk—each promoted model becomes the baseline for all later comparisons—and name it the "KEEP ratchet" [121].

F5: Where transfer is measured, selected gains shrink, sometimes to zero. Two core systems report an explicit held-out check of their candidates. In EvoSkillRec, five candidates lie on the validation Pareto front, "but only P1 (round 19) retains both improvements on the test set" [48]; P1, the front's highest-validation candidate, is also the one that keeps both gains. In Population Evolution, a ranking candidate gains 10.66% on a peer dataset and loses 0.79%

Table 3: Evaluation audit (full text, 22 October 2026). C1 metric: Full = full-catalogue ranking; Sampled = ranking against sampled negatives; Rank? = ranking metric with the candidate set not stated; AUC = AUC/logloss/loss proxy; A/B = online only; other = a non-recommendation metric; ? = no metric stated. C2 selection signal. C8 baselines: prod. = production model; pub. = published anchors. C4: P = repeats without spread, repeats in an ablation only, or a re-run rule. C9: P = part of the loop constrained (fixed evaluation code, gates, whitelists, sandboxing); G = metric-winning bugs, metric inflation or target leakage by the loop reported. Y = yes, N = no, P = partial, NS = not stated, – = not applicable.

System	Level	C1 metric	C2 select	C3 test once	C4 repeats	C5 gap	C6 null	C8 baselines	C9 isolation	A/B	Code
Self-EvolveRec [39]	L2	Sampled	val	NS	N	P	P	N	NS	N	Y
RecHarness [55]	L2	Sampled	val	Y	P	N	P	NS	P	Y	Y
Google/YouTube [95]	L1+L2	AUC	val	NS	P	P	N	prod.	NS	Y	N
EvoRec [66]	L2	Rank?	NS	NS	N	N	P	NS	NS	Y	N
VirtualMLE [10]	L1	Rank?	val	Y	P	P	Y	Y	Y	N	Y
NOVA [60]	L2	AUC	NS	NS	N	N	Y	prod.	P	Y	N
AgentX [43]	L2+meta	A/B	A/B	–	N	P	N	prod.	P	Y	N
AgentX-Model [110]	L2	AUC	NS	NS	P	N	P	prod.	P	Y	N
Auto-RecSys [46]	L2+meta	?	–	–	N	N	N	prod.	NS	N	N
AutoLR [121]	L1+L2	?	NS	NS	N	N	N	prod.	P	Y	N
CORAL [4]	L1	A/B	A/B	–	N	N	N	prod.	–	Y	N
AgenticRecTune [102]	L1+meta	A/B	A/B	–	N	N	N	prod.	–	Y	N
Sortify [18]	L1	A/B	A/B	–	N	P	N	prod.	–	Y	N
GEARS [113]	L1	A/B	A/B	–	N	P	N	prod.	–	Y	N
RecEvolve [76]	L2	Rank?	NS	NS	N	N	N	prod.	G	Y	N
Astar [30]	L2+meta	Rank?	loss	NS	N	P	P	prod.	NS	Y	N
A-MLE [26]	L1+L2	AUC	NS	NS	P	N	P	prod.	P	N	N
SR-Agent [108]	L1	A/B	A/B	–	N	P	N	prod.	P	Y	N
EvoSkillRec [48]	L2	AUC	val	Y	N	Y	Y	P	P	N	Y
RankEvolve-HSTU [16]	L1+L2	Full	test	N	N	P	P	pub.	G	N	later
Population Evolution [78]	L2	Full	NS	P	N	Y	P	Y	P	N	later
Trajectories-to-Evidence [135]	L2	AUC	NS	NS	N	P	P	prod.	P	Y	N
EvoPilot [24]	L1+L2	Rank?	NS	NS	P	P	N	prod.	G	Y	N
Amazon AutoResearch [12]	L2+L3	Full	test	N	P	P	N	Y	G	N	N
PILOT [52]	L1	A/B	A/B	–	N	P	N	prod.	P	Y	N
A/B Agent [35]	L1+meta	A/B	A/B	–	N	N	N	prod.	P	Y	N
DMRL [124]	L1+meta	A/B	A/B	–	N	P	N	pub.	P	Y	N
Comparison points outside recommendation											
EvoRank (LTR) [77]	L2	other	val	Y	Y	Y	Y	Y	Y	N	Y
AI Co-Scientist (search) [101]	L2	AUC	test	N	N	N	N	P	G	N	N
RankEvolve-IR (lexical) [72]	L2	Full	NS	Y	N	Y	P	N	N	N	Y
Evolving Semantic Indexes (GR) [71]	L3	Full	val	Y	P	Y	N	seed	Y	N	N
AgentLTV (LTV) [99]	L2	other	test	NS	N	N	P	NS	N	N	Y
Feedback Contract (CTR/LTR) [112]	L2	AUC	val	Y	Y	Y	N	seed	Y	N	N
PEAR (search) [97]	L1	A/B	A/B	–	P	P	N	prod.	NS	Y	N

at larger scale, and two of three peer-screened designs keep their gains; sharing evidence across searches raises the larger-scale pass rate for ranking from 11/60 to 18/60 [78]. Amazon's peak lift of 1.82× lies "~2.6σ above the across-date mean" of 1.63× [12], and EvoPilot's offline gain of 3.20 points became an online gain of 0.66% on a different metric and population [24]. Kuaishou's trajectory analysis finds that for 22 of 30 methods the last round falls below an earlier best [135], so the final state of a loop is not necessarily its best. The comparison points tell the same story: in EvoRank, the gains of the first (loss-evolution) campaign "were almost entirely fitness noise" on 60k held-out queries, and about 70% of the second campaign's gains survived [77]; in Evolving Semantic Indexes, a proxy decoder ranks semantic-ID programs well (Spearman ρ = 0.90), but once the generator's training objective may change, "the proxy becomes not only less informative but misleading", with five programs ranked inversely on proxy and target ($\rho = -0.80$, $n = 5$) [71]; in an e-commerce search loop, effects on a shadow-traffic proxy were several to tens of times larger than in decision-grade online experiments, and five promoted candidates showed no significant gain [97]. The cleanest measurement is a controlled one: on a label-permuted task where no real gain exists, a coding agent given exact validation scores "ended by reporting a gain of 0.0157" [112].

F6: Null controls are rare. Three core systems compare against a non-LLM search at an equal or favourable budget: VirtualMLE (grid search and Bayesian optimisation), NOVA (Optuna-TPE, matched

evaluation budget but unmatched LLM cost, on one task) and EvoSkillRec (NASRec, alongside OpenEvolve); in all three the LLM loop is reported to win. Nine have partial controls (unmatched budgets, a random arm router that still uses LLM edits, cross-LLM swaps, or replay-only comparisons), and fifteen have none. No core system runs an equal-budget random search over its own edit space. Where random selection was run, it was competitive: in AgentX-Model's historical replay, which randomises the choice of experiments rather than the edits, in Scenario A "Uniform Random's mean exceeds Fixed, Greedy, and Joint" policies, and no policy led consistently across environments [110], and in EvoRank's loss campaign an equal-budget random search transferred as well or better ($p = 0.03$) [77]. AutoLR states plainly that it "does not establish superiority over random, greedy, or direct LLM-based routing under a matched trial budget" [121]. Outside recommendation, classical optimisers beat LLM agents in fixed hyperparameter spaces [23, 85], which is why the null control matters.

F7: Agents find shortcuts in the evaluation; humans and monitoring, not the acceptance rule, catch them. Four core systems and one comparison point report metric-winning bugs, metric inflation or target leakage produced by the loop itself. RecEvolve's agent "discovered that drastically reducing the batch size (from 8k down to 1k) artificially inflated proxy metrics"; a human caught it and rolled it back [76]. On HSTU, RankEvolve's agents produced a genre feature that leaked the target (training loss collapsed to zero while evaluation spiked), DPO preference pairs mined at the held-out target position (a training-overlap flaw), and a naive multi-position evaluation that would have inflated NDCG@10 from 0.2192 to 0.2517/0.2577. The genre and DPO bugs surfaced as anomalous trajectories on the leaderboard, the multi-position inflation was reasoned out, and none was reported as a gain, but the best agent pair still had a 10.4% silent critical-defect rate on ExecML-HSTU, and a single Claude Code pass 35.4% [16]. In EvoPilot, "a favorable result used an evaluation date inside the candidate model's training window, so the result was retracted"; the same campaign recorded edits that history rewriting had silently removed and an evaluator whose aggregate metric "changed meaning" [24]. In Amazon's semantic-ID loop, "maximizing codebook size trivially improves weighted coherence", and the resulting hierarchy was unusable [12]. Agent reliability is a related problem: A-MLE's agent sometimes promoted a candidate on a single seed, which its automatic rerun rule materially reduced, and weaker models "pretend to have completed tasks" [26]. Prevention works when it is designed in: EvoRank's pre-search audit caught three target-encoding leakage bugs in the authors' own hand-built candidate before any search used it [77]. Among the borderline systems, AURA's agent fabricated 904 session identifiers in its diagnostic stage [38], and RecSys Factory's diagnostics uncovered a pre-existing upstream join bug that had "silently corrupted all downstream training samples" [3], a reminder that the data feeding the loop needs checks too. Only VirtualMLE states that the test set is hidden from the agent and the data pipeline immutable; thirteen systems constrain part of the loop (verification gates, whitelists, fixed evaluation code, frozen evaluation logic, isolated workspaces, sandboxed modules, bounded or legal-command actions), and five say nothing about it.

F8: Baselines, budgets and openness. The largest relative gains are measured against weak starting points: RecHarness improves average HR@10 by 85.85% over a cold-start GRU4Rec template and by 12.58% over HSTU, and also lists published numbers well above its cold-start base [55]; Self-EvolveRec configures all models with the same fixed hyperparameters (embedding 50, batch 128, learning rate 0.001) and does not describe tuning them [39]. Only VirtualMLE, Population Evolution and Amazon's AutoResearch (a hand-tuned baseline from four engineer-weeks) describe tuning their baselines. Budgets are reported in incompatible units (iterations, GPU-seconds, GPU-hours, ideas, tokens, dollars, yuan); twelve of 27 report LLM tokens or cost. Four of 27 release code, two promise to, and six do not name the LLM they use.

Online gates. Nineteen of the 27 systems report online experiments. These are strong evidence when well designed, and they are the main safeguard of the industrial loops. Nine state the experiment duration and nine report significance or confidence intervals; none corrects for multiple testing across the many rounds a loop runs (AutoLR discusses the problem but applies no correction). AutoLR's summed launch-review lifts are, in its own words, "descriptive and are not pooled treatment effects" [121]; Sortify chose its frozen round on the same online signal, then re-tested it in a fresh seven-day experiment [18], a practice we recommend. PILOT goes further in design: its final confirmation is "a single pre-registered test" that "runs once", because "the data collected while searching … cannot also serve as proof" [52]; the report does not yet include that confirmation, and its headline compares the best of five buckets before and after, "not randomized". Online selection has its own winner's curse: the arm chosen as best overstates its effect [7]. Online noise is visible inside single papers: DMRL's default configuration (Qwen3-8B, eight rollouts) appears in three tables run in "non-overlapping time windows" with LTV lifts of +0.052%, +0.064% and +0.050%, a spread of half its 0.028-point lead over the best baseline [124].

Does selection really overfit? A meta-analysis of 120 Kaggle competitions found little overfitting to public leaderboards [86], which cautions against assuming that reuse of a validation set must inflate results. The setting differs in ways that matter: Kaggle caps the rate of resubmission [9] and its leaderboards are probed by many teams with different model families [86], whereas an agent loop evaluates hundreds of correlated candidates on one split without a cap, keeps the maximum, and controls the code that produces the score. The direct measurements we found (F5) show that some selected gains shrink or vanish on transfer, while others survive. The safe conclusion is not that self-evolving recommenders do not work, but that the published evidence cannot yet tell how much of their gain is real.

7 A Reporting Standard and a Shared Protocol

The audit suggests that most of the problems are cheap to report; they are unreported or unenforced. We propose a reporting standard (Table 4) and a protocol that enforces the parts of it that can be enforced in software.

Cost of compliance. The expensive item is R4, which multiplies the cost of a session. Published per-session costs suggest it is affordable for academic settings: VirtualMLE reports $1.5–4.8 of LLM cost

Table 4: Reporting standard for self-evolving recommenders.

R1	Report full-catalogue ranking metrics (sampled metrics only as secondary), with the candidate set stated.
R2	Name the split or signal that accepts candidates; keep the test split out of every decision.
R3	Evaluate the finally selected candidate on test once, and report its validation score next to it.
R4	Repeat the whole evolution session ($\geq$ 5 seeds, or as many as affordable) and report every session's final test score with the median and range.
R5	Run a null control at equal trial and GPU budget: random search or HPO over the same configuration or edit-arm space, or, for open-ended edits, the same proposer with candidates accepted at random.
R6	Report the budget as a card: number of trials, GPU time, LLM tokens and cost, wall-clock, and the cost of training a learned proposer.
R7	Start from and compare against tuned baselines; report how they were tuned.
R8	State what the agent can read and write: data, labels, evaluation code; report gaming incidents found.
R9	For online gates: duration, traffic share, pre-specified metric and guardrails, confidence intervals, the number of rounds tested with a correction rule across them, and a fresh pre-registered confirmation of the final candidate.

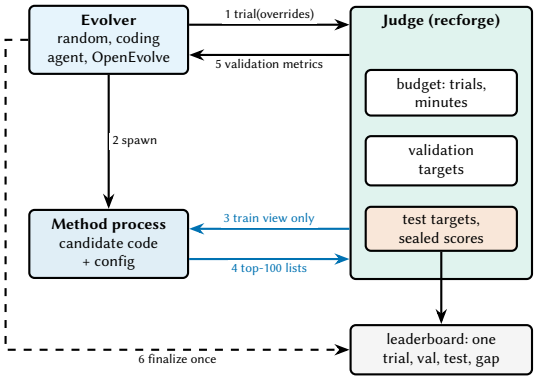


Figure 4: An evolve session in recforge. The evolver requests trials (1); each trial runs the candidate in its own process (2), which receives a training view without targets (3) and returns ranked lists (4). Only validation metrics flow back (5); test scores are computed and sealed. One `finalize` publishes a single trial with its validation score, test score, gap and budget (6). The seal holds at the API level; file-system isolation is planned.

per session [10], EvoRank about ten dollars for fifty iterations [77], and NOVA about $29 per task [60]; training cost dominates and is shared with R5. For industrial loops whose candidates take hundreds of GPU-hours [46], R4 can be applied to a replay or a reduced-scale proxy, as AgentX-Model does for scheduling policies [110].

recforge. We built recforge [82], an open benchmark that implements R1–R3, R5, part of R6, and part of R8 for sequential recommendation (Figure 4).

Its design choices follow from the audit:

- *One protocol.* Leave-one-out per user (last item test, second-to-last validation), full-catalogue ranking with the user's history removed, and HR and NDCG at 10 and 20 and MRR@100. Results from different protocol versions are never put in one table, and every result is keyed by the source hash of the method, its configuration and the code commit.
- *A sealed test at the API level.* Method code runs in its own process and receives only a training view of the data; asking it for a validation or test target raises an error, and the method returns ranked lists that the parent process scores. Sealed scores and session state are ordinary files, so an agent with file-system access could read test scores or alter the budget; until the planned container mode, the seal holds only against code that uses the API.
- *Budgeted evolve sessions.* A session has a trial and time budget. Each trial returns validation metrics only; test metrics are computed and sealed. A single `finalize` reveals the test metrics of exactly one trial, publishes it to the leaderboard with the session's budget attached, and closes the session.
- *Evolvers on equal terms.* Any evolver drives a session through the same interface: an equal-budget random search (the R5 null control), a wrapper that runs an external coding agent such as Claude Code or Codex, and an OpenEvolve adapter. Each run reports the validation-to-test gap of its finalized trial. These evolvers are implemented and tested on synthetic data but not yet exercised in this survey.
- *Baselines and data.* Popularity, ItemKNN, BPR, SASRec [37], HSTU [115] and TIGER [81] are built in, with MovieLens-1M, Amazon 2014 Beauty/Sports/Toys, MIND news, and the Kuaishou OpenOneRec release [131].

recforge does not yet satisfy the rest of the standard, and we list the gaps rather than claim them: the training view protects the API but not the file system, so code that opens the raw data files or the sealed score files can still read targets or test results (a container mode is planned); LLM tokens and cost are not recorded; repeated sessions are not aggregated automatically; there is no timeline-aware split [34]; and the OpenEvolve adapter has not yet been run end to end. We report no evolver results in this survey.

8 Open Problems

Isolation without crippling the agent. Code-level agents need to read data and change training code, and that is exactly the access that lets them leak targets or change what the metric measures (F7). Verification gates (NOVA), fixed evaluation code (AgentX-Model), immutable pipelines (VirtualMLE) and cross-agent review (RankEvolve) each close part of the gap. A principled design would give the agent a process that physically cannot reach evaluation labels or code, and would audit every metric-relevant change, such as batch size for in-batch metrics.

Proxies for expensive candidates. Industrial candidates take hours to hundreds of GPU-hours, so loops screen on proxies: small decoders [71], data subsets and peer datasets [78], training-loss curves and learned reward models [30]. The evidence shows that proxy fidelity depends on what is evolved. Characterising when a proxy

can be trusted, and spending budget to re-check proxies during search, is open.

Budget-normalised comparison of evolvers. There is no common unit of cost and no common task, so the field cannot say whether bandit routing, island evolution, research-agent pairs or learned proposers are better. A shared benchmark with a budget card (R6) and an equal-budget random baseline (R5) would allow the first such comparison.

Evolving generative recommenders. Semantic-ID construction, codebook collisions, constrained decoding and preference-alignment losses are new design dimensions with little human prior and large effects. They are natural targets for code-level evolution, and almost untouched (§5).

From offline wins to online value. RecEvolve observes that offline evaluations often show "win-win" results while online tests show trade-offs [76]. Loops that select on offline metrics inherit the gap; loops that select on online experiments face multiple testing and slow iteration. Combining the two with a fresh online confirmation of the final candidate, as Sortify does with a seven-day re-test, ideally pre-registered, is a starting point.

Circularity at the policy level. L4 systems are often judged by simulators from the same model family as the agent being improved. Held-out simulators [39], verifiable rewards [68] and reward-model benchmarks [114] are partial answers.

9 Threats to Validity

Coverage. The field is months old and grows weekly; our cut-off is 22 October 2026. Many systems are arXiv preprints whose versions change; we coded the version available on that date. Generic keyword search performed poorly, so coverage relies on citation-graph search and may miss systems that neither cite nor are cited by the core set.

Coding. The audit was coded from full text with verbatim evidence and then checked against the original papers by two adversarial readings: the first corrected 12 cells in nine rows, the second re-verified every cited sentence against its source and corrected the remaining cells and claims (both reports are released). The seven rows added by the second literature sweep were coded from the full text by one reading, and their quotes were rechecked against the source. We have not yet measured agreement between two independent coders. "Not stated" does not mean "not done": authors may have followed a practice without reporting it. We release the evidence so that authors and readers can correct individual cells, and we will update the table.

Industrial systems. Offline hygiene criteria are less important for systems whose final gate is a well-designed online experiment. We therefore report online-gate design separately (R9) and do not treat a missing offline test split as a defect when no offline claim is made.

Conflict of interest. The authors built recforge, the benchmark proposed in §7. recforge was developed alongside this survey, so its design reflects our reading of the literature; we list its gaps next to its features, and the audit itself does not depend on it.

10 Conclusion

Self-evolving recommenders moved in one year from an idea to production systems at many large platforms. The loops are inventive, and several report online gains under well-designed experiments. The offline evidence is weaker than the headlines suggest: selection splits are mostly unstated, test-once and repeated sessions are rarely reported, equal-budget random search is absent, and the few papers that check transfer find that some selected gains do not survive. Agents also find shortcuts in the evaluation, which humans and monitoring rather than the acceptance rules catch. Most of this is cheap to report. We hope the reporting standard in Table 4, the released audit data, and an open protocol such as recforge make it the norm, so that the next comparison of evolvers can be made on equal terms.

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